

AR Test: Channel First Stages & the Baseline Screen

May 21, 2026

Tasks done

- ▶ **Which political office moves local economic composition?** Instead of using Mayor-crossed channels by assumption, I verify which of them actually move sector shares.
- ▶ **Updated version of the AR test.** The AR test now includes an instrumented version of volume controls and is also evaluated for sector by size groups.

Summary of findings

- ▶ **Sector-share first stage.** At BNDES aggregate sector every channel's instrument block moves only the Infrastructure share (SW F from 10 to 33); the Agriculture and Industry shares are never identified. At the finer 12-group margin relevance is weak.
- ▶ **Volume first stage.** Mayor and Mayor · Governor move the volume control.
- ▶ **AR test, policy block.** Stack carrying the Governor block tend to reject the null of no composition effect, and the rejection survives instrumenting volume.
- ▶ **AR test, 12 groups.** Governor, President, and Governor · President reject, but on weak first stages — read as a possible exclusion/direct-effect warning, not as a composition-channel finding.

Sector shares: relevance concentrates in Infrastructure

First stage of the endogenous sector employment shares — BNDES aggregate sector (Services is the omitted base). All 18 evaluated stacks; each cell is the cluster-robust SW first-stage F per sector share.

Singletons and Mayor stacks

Stack	SW F per sector share			KP
	Agriculture	Industry	Infrastructure	
{M}	1.44	0.67	6.10	7915.41
{G}	7.93	2.60	19.39	623.28
{P}	0.64	1.15	32.66	47985.06
{M·G}	3.13	1.58	10.17	326.26
{M·P}	1.97	2.17	14.25	14286.03
{G·P}	1.29	0.94	23.85	280.19
{M·G·P}	1.28	1.36	11.63	25705.82
{M, M·P}	1.55	1.57	8.18	6848.51
{M, M·G}	3.28	1.10	6.13	677.95
{M, M·G·P}	1.41	1.17	8.47	4080.36
{M, M·P, M·G, M·G·P}	2.26	1.07	8.14	5700.49

Parent pairs, parent + interaction

Stack	SW F per sector share			KP
	Agriculture	Industry	Infrastructure	
{M, G}	4.99	1.91	11.03	4967.88
{M, P}	1.00	1.37	16.58	32578.28
{G, P}	4.19	1.75	20.13	29.16
{M, P, M·P}	1.83	1.62	11.16	44791.43
{M, G, M·G}	4.22	1.79	7.26	4507.92
{G, P, G·P}	3.08	1.32	18.72	159.14
{M, G, P, M·G·P}	2.86	1.35	11.31	126.75

Relevance concentrates in the Infrastructure share across singletons and stacks alike; Agriculture and Industry rarely clear the threshold. KP is the Kleibergen–Paap rank statistic. The first stage matches the AR test — same unit, FE, exposure control, and clustering.

Twelve groups: sector-share relevance is weak

First stage of the endogenous sector shares — BNDES aggregate sector $\times$ firm size (12 groups). All 18 evaluated stacks; per stack the KP rank statistic, the SW F range over the 11 shares, and the count clearing SW $F \geq 10$.

Singletons and Mayor stacks

Stack	KP	SW F min	SW F max	Identified shares
{M}	10.40	0.75	6.95	0/11
{G}	1.77	0.96	13.22	3/11
{P}	1.72	1.55	21.64	2/11
{M·G}	50.65	0.80	7.75	0/11
{M·P}	10.85	1.39	9.65	0/11
{G·P}	142.00	0.94	12.49	1/11
{M·G·P}	88.79	1.12	7.86	0/11
{M, M·P}	2979.63	1.24	6.29	0/11
{M, M·G}	5967.04	1.12	4.72	0/11
{M, M·G·P}	62.98	1.05	6.77	0/11
{M, M·P, M·G, M·G·P}	1042.46	1.22	4.69	0/11

Parent pairs, parent + interaction

Stack	KP	SW F min	SW F max	Identified shares
{M, G}	532.48	1.13	8.06	0/11
{M, P}	32.31	1.21	10.84	1/11
{G, P}	3.85	1.40	12.27	1/11
{M, P, M·P}	649.80	1.03	7.97	0/11
{M, G, M·G}	806.32	1.23	4.93	0/11
{G, P, G·P}	172.86	1.12	10.76	1/11
{M, G, P, M·G·P}	6615.08	1.12	6.73	0/11

Relevance is thin at the 12-group margin — at most 3 of 11 shares are identified, and most stacks identify none. The KP statistic and the SW F disagree. The first stage matches the AR test — same unit, FE, exposure control, and clustering.

Volume control: Mayor and Mayor · Governor move volume

First stage of the volume control — each channel's aggregate instrument regressed on the volume ratio.

Channel	Coefficient	Partial F	p -value	Volume instrument
Mayor	-0.397	3.97**	0.046	Yes
Governor	0.325	1.85	0.174	No
President	-0.188	1.27	0.261	No
Mayor · Governor	-0.77	4.60**	0.032	Yes
Mayor · President	-0.0845	0.13	0.724	No
Governor · President	-0.187	0.25	0.616	No
Mayor · Gov. · President	-0.149	0.23	0.630	No

Mayor and Mayor · Governor clear the 5% relevance gate; Governor, President, and the remaining channels do not. These two channels supply the instrument for the volume ratio in the volume-instrumented AR specification. The partial F is a cluster-robust Wald test on the channel's aggregate instrument; municipality-year observations, muni + year FE, clustered by municipality, exposure control included throughout. The 12-group margin selects the same two channels. *

$p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

BNDES aggregate sector: Governor-containing stacks reject

AR test by excluded-instrument stack. {M, P, M · P} means M, P, and M · P are all excluded instruments at once.

Stack	K_Z	AR F (no vol.)	AR F (vol. control)	AR F (vol. instr.)	Relevant shares
{M·P}	4	1.311 [0.263]	1.311 [0.263]	0.435 [0.784]	1/3
{M·G}	4	3.192** [0.012]	3.195** [0.012]	0.296 [0.881]	1/3
{M·G·P}	4	1.378 [0.239]	1.378 [0.239]	0.953 [0.432]	1/3
{G}	4	11.962*** [<0.001]	11.965*** [<0.001]	11.980*** [<0.001]	1/3
{P}	4	0.919 [0.452]	0.919 [0.452]	0.886 [0.471]	1/3
{G·P}	4	1.974* [0.095]	1.974* [0.095]	1.600 [0.171]	1/3
{M, G}	8	6.519*** [<0.001]	6.520*** [<0.001]	6.457*** [<0.001]	1/3
{M, P}	8	1.164 [0.317]	1.164 [0.317]	0.930 [0.490]	1/3
{G, P}	8	6.630*** [<0.001]	6.632*** [<0.001]	6.674*** [<0.001]	1/3
{M, P, M·P}	12	1.142 [0.320]	1.142 [0.320]	1.026 [0.421]	1/3
{G, P, G·P}	12	5.230*** [<0.001]	5.232*** [<0.001]	5.196*** [<0.001]	1/3
{M, G, P, M·G·P}	16	4.209*** [<0.001]	4.209*** [<0.001]	4.128*** [<0.001]	1/3

Three volume treatments: no control, volume control, and volume instrumented (Full IV). The AR statistic is a cluster-robust joint Wald test on the instrument coefficients only. The exposure control is included in every specification. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Twelve groups: all three screened stacks reject, on weak first stages

AR test by excluded-instrument stack — BNDES aggregate sector $\times$ firm size (12 groups).

Stack	K_Z	AR F (no vol.)	AR F (vol. control)	AR F (vol. instr.)	Relevant shares
{G}	12	6.003*** [<0.001]	6.004*** [<0.001]	5.932*** [<0.001]	3/11
{P}	12	2.144** [0.012]	2.145** [0.012]	1.623* [0.078]	2/11
{G·P}	12	2.132** [0.012]	2.132** [0.012]	2.032** [0.018]	1/11

Governor, President, and Governor · President all reject without and with a volume control. Under Full IV Governor and Governor · President still reject, but President falls to the 10% margin. The first stages are weak (1 to 3 of 11 shares identified), so a rejection here is read as a possible exclusion/direct-effect warning, not as evidence of a composition channel. The AR statistic is a cluster-robust joint Wald test on the instrument coefficients only. The exposure control is included in every specification. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.